

Neural Networks Laboratory Manual

Lab 1: Single Layer Perceptron

Objective 1: - Implement a single-layer perceptron for binary classification (AND/OR gate) from scratch and visualize the decision boundaries.

Theory: Single layer perceptron and its importance.

Objective 2: - Implement a binary moon classification, "limitation test" - show that a single-layer perceptron cannot handle non-linear separation. Use scikit-learn moon generator.

Theory: single layer perceptron and its limitations.

Objective 3: - Implement a MLE and MAP estimation

Theory: MLE and MAP

Lab 2: Multilayer Perceptron (MLP)

Objective: - Implement a Backpropagation Algorithm from scratch. (Assume you have AND/XOR gate with two inputs, train a multilayer perceptron).

Theory: - Include theory of backpropagation

Lab 3: Linear Prediction

Objective: - Apply linear regression to predict house prices - Plot actual vs predicted.

Theory: - Model the output as a linear combination of input variables and evaluate the model's performance using appropriate metrics like MSE.

Tools: Scikit-learn, NumPy

Dataset: Boston Housing

Lab 4: Pattern Classification

Objective: - Use neural networks for pattern recognition

Theory: - Use MLP/CNN for image or signal classification

Tools: TensorFlow/Keras, Scikit-learn

Dataset: Digits or Fashion MNIST

Lab 5: Supervised Learning

Objective: - Train models using supervised learning with labeled data.

Theory: - The model is trained on input-output pairs to minimize a loss function.

Tools: Scikit-learn, Keras

Dataset: Cats vs Dogs or Breast Cancer or Wine dataset

Tasks: - Train classifier - Evaluate performance metrics

Lab 6: Unsupervised Learning

Objective: - Perform clustering using K-Means from scratch.

Theory: - Group similar data points together without using labels.

Tools: Scikit-learn

Dataset: Iris or synthetic dataset

Tasks: - Visualize clusters - Evaluate inertia/score

Lab 7: Recurrent Neural Network (RNN)

Objective: - Predict sequences using RNN.

Theory: - RNNs use feedback connections to process sequences

Tools: TensorFlow/Keras

Dataset: Synthetic sine wave/time-series

Tasks: - Predict next number in sequence

Lab Report Format

1. Title
2. Objective
3. Theory
4. Algorithm
5. Code
6. Results

7. Conclusion